

NAS JIAXING
AFTER-SCHOOL ECA

AI CLUB

Student Coursebook

Build, understand and use intelligent systems responsibly
搭建、理解并负责任地使用智能系统

AI ♦ 101

Grades 7-12

15 lessons × 60 minutes

Linux • PC hardware • AI agents • Local models • Responsible cloud AI

PRACTICAL STUDENT WORKBOOK

Name: _____ Class: _____
ECA term: _____

Course Overview / 课程概览

Welcome to AI Club! This is a supervised, hands-on after-school ECA. Across fifteen one-hour sessions, you will use Ubuntu Linux, inspect and safely upgrade practice computers, explore how AI models work, operate a teacher-approved OpenClaw environment, run small local models and make responsible choices about cloud AI.

欢迎加入AI俱乐部! 这是一项由老师监督、以实践为主的课后拓展活动。通过15节一小时课程,你将使用Ubuntu Linux,检查并安全升级练习电脑,了解AI模型的工作方式,操作经老师批准的OpenClaw环境,运行小型本地模型,并学习负责任地使用云端AI。

Important: Hardware, accounts, network access, model files and credentials are controlled by the instructor. Do not repeat installation or pairing steps on personal or school devices unless the instructor has approved that exact device.

重要: 硬件、账户、网络访问、模型文件和凭据均由老师管理。除非老师明确批准某一台设备,否则不要在个人或学校设备上重复安装或配对操作。

What to Bring

- ☐ A notebook and pen
- ☐ A USB drive (8GB+) to save your work / U盘 (8GB以上) 保存作业
- ☐ Curiosity!

What We Provide

- ☐ Laptop per student
- ☐ All software and course materials
- ☐ Demo hardware for PC building lessons

Course Pathway

1. Linux Foundations / Linux基础

Lessons 1-3 / 第1-3课

2. Hardware Lab

Lessons 4-5 / 第4-5课

3. AI Foundations / AI基础

Lessons 6-7 / 第6-7课

4. Agents + Local AI

Lessons 8-13 / 第8-13课

5. Safe Cloud + Showcase

Lessons 14-15 / 第14-15课

Safety & Classroom Rules / 安全与课堂守则

#	Rule	What it means in this ECA
1	Teacher approval first 先获得老师批准	Use only the assigned practice device, account, network and model. / 仅使用指定的练习设备、账户、网络和模型。
2	Electrical safety 用电安全	Shut down, switch off and unplug before opening a case. Never open a power-supply unit. / 打开机箱前关机、断电并拔线；绝不拆开电源内部。
3	ESD protection 静电防护	Use the instructor's ESD procedure and handle components by their edges. / 按老师要求做好静电防护，并握住部件边缘。
4	Protect secrets 保护秘密信息	Never copy passwords, API keys, bot tokens or setup codes into this book, screenshots or chat. / 不要把密码、API密钥、机器人令牌或配对码写进手册、截图或聊天。
5	Protect people and data 保护个人与数据	Do not upload names, faces, student work, private files or identifying details without permission. / 未经许可，不上传姓名、面部、学生作业、私密文件或身份信息。
6	Think and verify 思考并核实	AI output is a draft or prediction. Check important claims and explain how AI helped. / AI输出是草稿或预测；核实重要信息，并说明AI如何提供帮助。
7	Stop when uncertain 不确定就停下	Do not force hardware, approve unknown access or run a repair command you do not understand. / 不要硬插硬拆、批准未知访问或运行不理解的修复命令。

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Starting-Point Skills Passport

起点技能护照

This is not a test. Circle one number for each statement: 1 = new to me, 4 = I can explain or demonstrate it.

这不是考试。每项圈出一个数字：1=刚接触，4=我能解释或演示。

My starting confidence / 我的起点信心

I can... / 我能……	Start (1-4)
work safely with a computer and identify key components / 安全使用电脑并识别主要部件	1 2 3 4
navigate Linux and manage practice files in the terminal / 使用Linux并在终端中管理练习文件	1 2 3 4
explain how an AI model learns patterns and why it can make mistakes / 解释AI模型如何学习模式及为何会出错	1 2 3 4
test an AI answer using evidence instead of trusting it automatically / 用证据检验AI回答，而不是自动相信	1 2 3 4
protect personal data, passwords, tokens and school information / 保护个人数据、密码、令牌和学校信息	1 2 3 4
run and compare an approved local AI model / 运行并比较经批准的本地AI模型	1 2 3 4
use OpenClaw status and diagnostic commands safely / 安全使用OpenClaw状态和诊断命令	1 2 3 4
plan, test and explain a small responsible-AI project / 规划、测试并解释一个负责任AI小项目	1 2 3 4

My goal for this club / 我的课程目标

I would like to learn how to... / 我想学会……

One question I already have / 我现在已经有的一个问题： _____

Lesson 1

Ubuntu Installation & Desktop Basics / Ubuntu安装与桌面环境入门

Suggested timing: 8 + 25 + 20 + 7 = 60 min

Learning Objectives

- Install Ubuntu 24.04 LTS from a bootable USB
- Navigate the GNOME desktop, settings, and file manager
- Create your first user account and understand permissions

Key Vocabulary

English	中文	Definition
OS	操作系统	Software managing computer hardware
Distribution	发行版	A packaged version of Linux
Bootable USB	可启动U盘	USB that can start a computer
Partition	分区	Section of a hard drive
GNOME	GNOME桌面	Default Ubuntu desktop environment
sudo	超级用户权限	Run a command with administrator privileges

STARTER

"Linux Around the World" — In pairs, list every device or service you used today that runs on Linux (Android phones, servers, smart TVs, ATMs, etc.). Share with the class. Write three examples.

"Linux无处不在" — 两人一组，列出你今天使用的、运行在Linux上的设备或服务。与全班分享。在笔记本写下三个例子。

Worksheet

My three Linux-powered things / 我的三个Linux设备：

1. _____
2. _____
3. _____

MAIN 1

Installing Ubuntu on an assigned practice laptop — Use only the laptop labelled by your instructor. The class image is Ubuntu 24.04 LTS or another school-approved LTS image. Work with a partner and pause at every STOP point.

在指定练习笔记本上安装Ubuntu — 仅使用老师指定并贴有标签的电脑。课堂镜像为Ubuntu 24.04 LTS或其他经学校批准的LTS版本。两人合作，并在每个“停止”处暂停。

STOP / 停止：Installing an operating system can erase every file on a disk. The instructor must confirm the device and target disk before you continue. / 安装操作系统可能清除磁盘上的全部文件。继续前必须由老师确认设备和目标磁盘。

1. Insert the instructor-prepared USB and power on.
2. Open the one-time boot menu (often F12, Esc or F2).
3. Select the labelled USB entry.
4. Choose "Try or Install Ubuntu".
5. At the live desktop, raise your hand. ← STOP
6. Select language and keyboard layout with the instructor.
7. At "Installation type" and disk selection, raise your hand. ← STOP
8. Use the username written on the device label. Create a private password.
9. Complete installation; remove the USB only when instructed.
10. After restart, check the release:
`lsb_release -ds`

Installation record

Practice device label / 练习设备编号: _____
Ubuntu release / Ubuntu版本: _____
Teacher disk check completed / 老师已确认磁盘: ☐ Yes / 是
Do not write your password here. / 不要在此处写密码。

MAIN 2

Desktop Exploration Scavenger Hunt — Explore the Ubuntu desktop and complete the checklist.

桌面探索寻宝 — 探索Ubuntu桌面并完成以下检查清单。

- ☐ Open Files and create folder AI_Club in Home
- ☐ Open Settings → Wi-Fi and connect
- ☐ Change resolution to 1920×1080
- ☐ Open Firefox, visit ubuntu.com, bookmark it
- ☐ Take screenshot (PrtSc) save to AI_Club/screenshot1.png
- ☐ Open Terminal from app grid

```
mkdir ~/AI_Club
```

Worksheet

Screenshot saved? Yes / No / 截图已保存? 是 / 否 _____

Terminal opened? Yes / No / 终端已打开? 是 / 否 _____

REFLECT

"My First Linux Moment" — Write 2–3 sentences: What surprised you about Ubuntu? What was confusing? What do you want to learn next?

"我的首个Linux时刻" — 写2–3句话: Ubuntu的哪一点让你惊讶? 哪里让你困惑? 接下来想学什么?

Reflection Space

VISUAL SUMMARY

Ubuntu desktop map / Ubuntu桌面导航

Activities

Find apps and windows / 查找应用与窗口

Files

Home → AI_Club → Lessons / 主目录→课程文件夹

Settings

Network, display, users / 网络、显示、用户

Terminal

Type commands carefully / 谨慎输入命令

STOP before disk changes → teacher checks the device / 更改磁盘前停下→由老师确认设备

Lesson 2

Terminal Commands & File System / 终端命令与文件系统管理

Suggested timing: 7 + 20 + 25 + 8 = 60 min

Learning Objectives

- Use 15+ essential terminal commands
- Understand the Linux directory tree structure
- Create, move, copy, and delete files and folders

Key Vocabulary

English	中文	Definition
Terminal	终端	Command-line interface
Shell (Bash)	Shell (Bash)	Command interpreter
Directory	目录	Folder in Linux
Path	路径	Location of a file/folder
Absolute path	绝对路径	Full path from root
Relative path	相对路径	Path from current location
Root (/)	根目录	Top-level directory
Home (~)	主目录	Your user folder

STARTER

"Terminal vs GUI Race" — In pairs, one uses GUI (Files app + mouse), the other uses Terminal. Race to: (1) create 3 folders, (2) create a text file inside each, (3) delete one folder. Which was faster? Discuss why terminal power matters.

"终端与图形界面对决" — 两人一组，一位用图形界面，另一位用终端。比赛完成：(1) 创建3个文件夹，(2) 每个文件夹内创建文本文件，(3) 删除一个文件夹。哪个更快？讨论为什么终端更强大。

Worksheet

GUI time / 图形界面用时：_____ seconds / 秒

Terminal time / 终端用时：_____ seconds / 秒

Winner / 获胜方：_____

MAIN 1

Linux filesystem treasure hunt — Predict what each command will do, run it, then check the result. All changes stay inside ~/AI_Club/sandbox.

Linux文件系统寻宝 — 先预测每条命令的作用，再运行并检查结果。所有更改仅在~/AI_Club/sandbox中进行。

```
pwd # show current location / 显示当前位置
mkdir -p ~/AI_Club/sandbox
cd ~/AI_Club/sandbox
touch notes.txt todo.txt
mkdir Lesson2
mv notes.txt Lesson2/
cp todo.txt Lesson2/todo_backup.txt
echo "Hello AI Club" > hello.txt
cat hello.txt
tree ~/AI_Club/sandbox
rm -i todo.txt # confirm one practice-file deletion / 确认删除练习文件
```

Never add sudo to a delete command and never delete outside the assigned practice folder. / 不要在删除命令前加sudo，也不要删除指定练习文件夹之外的内容。

Predict → Run → Check

Command	My prediction	What happened?
mkdir	_____	_____
mv	_____	_____
cp	_____	_____
rm -i	_____	_____

MAIN 2

Build Your Directory Structure — Create the exact folder and file structure shown below using ONLY terminal commands. Verify with tree.

构建你的目录结构 — 使用纯终端命令创建下图所示结构。用tree验证。

```
AI_Club/  
|-- Lessons/  
|   |-- Lesson01/  
|       |-- summary.txt  
|   |-- Lesson02/  
|       |-- commands.txt  
|-- Projects/  
|   |-- my_first_ai.txt  
+-- Resources/  
    |-- links.txt
```

```
mkdir -p ~/AI_Club/Lessons/Lesson01  
touch ~/AI_Club/Lessons/Lesson01/summary.txt  
# ... continue for all folders and files  
  
# Verify:  
tree ~/AI_Club
```

Worksheet

Draw your tree output / 画出你的tree输出:

REFLECT

"Terminal Confidence Check" — Rate your confidence 1–5 for each skill. Write one command you will definitely remember.

"终端信心检查" — 为每项技能打分1–5。写出一个你肯定会记住的命令。

Reflection Space

Skill	1–5
Navigating directories	_____
Creating files	_____
Moving/copying	_____
Using sudo safely	_____

My never-forget command / 我绝不会忘的命令： _____

VISUAL SUMMARY

A safe command pattern

1 LOOK

pwd • ls • tree

2 PLAN

Say what the command will change / 先说明将更改什么

3 ACT

mkdir • touch • cp • mv

4 CHECK

ls -la • cat • tree

Use rm -i only inside the practice folder / 仅在练习文件夹内使用 rm -i

Lesson 2 Notes

第2课笔记

Use this page for commands, test evidence, diagrams or questions to revisit. / 可记录命令、测试证据、图示或待解决问题。

Lesson 3

Productivity Apps & System Tools / 实用工具与生产力软件配置

Suggested timing: 7 + 20 + 25 + 8 = 60 min

Learning Objectives

- Explain the difference between repositories, packages and app stores
- Install one teacher-approved package using apt
- Use system tools to inspect CPU, memory and storage

Key Vocabulary

English	中文	Definition
Repository	软件源	Software storage location
Package manager	包管理器	Tool to install software
IDE	集成开发环境	Integrated Development Environment
Extension	插件/扩展	Add-on for software
Theme	主题	Visual style

STARTER

"App Matchmaker" — Match the task to the best app: (1) Write Python code, (2) Edit a spreadsheet, (3) Take screenshots, (4) Monitor CPU usage. Apps: VS Code, LibreOffice Calc, Flameshot, htop.

"软件配对" — 将任务与最佳软件配对: (1) 编写Python代码, (2) 编辑电子表格, (3) 截图, (4) 监控CPU使用率。

Worksheet

1. Write Python / 写Python → _____
2. Spreadsheet / 电子表格 → _____
3. Screenshot / 截图 → _____
4. CPU monitor / CPU监控 → _____

MAIN 1

Install one small approved toolkit — Read each command aloud before running it. The instructor may use a local package cache if the network is slow.

安装一个小型批准工具包 — 运行前先说出每条命令的作用。网络较慢时, 老师可能使用本地软件缓存。

```
sudo apt update
apt show tree
sudo apt install -y tree htop
tree --version
htop
```

VS Code, LibreOffice and screenshot tools should be installed only from the school-approved Software/App Centre or package source. Do not paste third-party installation commands from search results.

VS Code、LibreOffice和截图工具仅应通过学校批准的软件中心或软件源安装。不要粘贴搜索结果中的第三方安装命令。

Source check

Package installed / 已安装软件包: _____ Publisher/source / 发布者或来源: _____

What permission did sudo grant? / sudo授予了什么权限?

MAIN 2

System monitoring challenge — Inspect your machine without changing its configuration. Record evidence, then close htop with q or F10.

系统监控挑战 — 在不更改配置的情况下检查电脑。记录证据，然后按q或F10退出htop。

```
lscpu | grep -E "Model name|CPU(s)"
free -h
lsblk -o NAME,SIZE,TYPE,MOUNTPOINTS
htop
```

Hardware snapshot

CPU model / CPU型号: _____

Total RAM / 内存总量: _____ Storage / 存储: _____

One process using CPU / 一个占用CPU的进程: _____

REFLECT

"My Ubuntu Setup" — List three apps you installed and one customization. Would you use this setup at home? Why or why not?

"我的Ubuntu配置" — 列出今天安装的两个软件项目和一项自定义设置。你会在家里的电脑上使用这套配置吗？

Reflection Space

App 1 / 软件1: _____

App 2 / 软件2: _____

App 3 / 软件3: _____

Customization / 自定义: _____

Home use? Yes/No because / 家用? 是/否, 因为: _____

VISUAL SUMMARY

Where software comes from

APT

Ubuntu repositories / Ubuntu软件源

App Centre

Visual installation / 图形化安装

Snap / Flatpak

Packaged apps with different update models / 不同更新方式

Teacher-approved source

Check publisher and permissions / 核对发布者与权限

Never paste an installation command you do not understand / 不要粘贴自己不理解的安装命令

Lesson 4

PC Hardware Identification & Functions / 电脑硬件识别与功能原理

Suggested timing: 7 + 20 + 25 + 8 = 60 min

Learning Objectives

- Identify and explain the function of 8+ core PC components
- Read specification labels and model numbers
- Use lshw, lspci, and lsusb to inspect hardware via software

Key Vocabulary

English	中文	Definition
CPU	中央处理器	The "brain" executing instructions
RAM	内存	Temporary fast storage for active tasks
Storage (SSD/HDD)	存储	Permanent data retention
Motherboard	主板	Main circuit board
GPU	显卡	Renders images; also used for AI
PSU	电源	Power Supply Unit
PCIe	PCIe插槽	Expansion slot for GPU, NVMe
Thermal Paste	硅脂	Compound between CPU and cooler

STARTER

"Hardware Detective" — Look at the demo PC on the teacher's desk. Without opening it, identify as many external ports and components as you can. Draw arrows on the worksheet.

"硬件侦探" — 观察讲台上的演示电脑。在不打开机箱的情况下，尽可能多地识别外部接口和部件。在练习页上画箭头标注。

Worksheet

Label these ports / 标注这些接口:

- ☐ USB-A ☐ USB-C ☐ HDMI ☐ DisplayPort
☐ Ethernet / 网线口 ☐ Audio Jacks / 音频接口 ☐ Power / 电源

[Draw the back of a PC and label / 画出电脑背面并标注]

MAIN 1

Software Hardware Audit — Use terminal commands to create a complete hardware report of your laptop. Save it to a file.

软件硬件审计 — 使用终端命令为笔记本电脑创建完整的硬件报告，并保存到文件。

```
sudo lshw > ~/AI_Club/hw_report.txt
lspci > ~/AI_Club/pci_devices.txt
lsusb > ~/AI_Club/usb_devices.txt
lscpu | grep "Model name" >> ~/AI_Club/hw_report.txt
free -h >> ~/AI_Club/hw_report.txt
lsblk >> ~/AI_Club/hw_report.txt

# Read your report
cat ~/AI_Club/hw_report.txt
```

Hardware Audit Summary

Component	Model/Size	Rating
CPU	_____	★★★★★

RAM	_____	★★★★★
Storage	_____	★★★★★
GPU	_____	★★★★★

MAIN 2

Component Matching Game — Your teacher has placed physical components on the table. Match each to its description card, then fill the function table.

部件配对游戏 — 老师已在桌上摆放实物部件。将每个部件与功能描述卡配对，然后填写功能表。

Component Function Table

Component	Primary Function	AI Relevance
CPU	Executes instructions	Runs AI inference
RAM	_____	Holds model data
GPU	_____	_____
SSD	_____	Stores AI models
PSU	_____	Powers AI workstations
Motherboard	_____	Connects AI accelerators

REFLECT

"The AI Hardware Dream" — If you could build a PC specifically for running local AI, which three components would you prioritise for upgrades and why?

"AI硬件梦想" — 如果你要组装一台专门用于运行本地AI的电脑，你会优先升级哪三个部件？为什么？

Reflection Space

1. _____ because / 因为：

2. _____ because / 因为：

3. _____ because / 因为：

VISUAL SUMMARY

How a computer handles a task

INPUT

Keyboard • files • network / 键盘 • 文件 • 网络

CPU + RAM

Process instructions + hold active data / 处理指令+保存活动数据

GPU (optional)

Parallel graphics/AI work / 并行图形与AI计算

STORAGE

Save programs and files / 保存程序与文件

Motherboard connects the parts; PSU supplies power / 主板连接部件；电源供电

Lesson 5

Safe Component Handling & Upgrades / 安全部件操作与升级实践

Suggested timing: 8 + 20 + 25 + 7 = 60 min

Learning Objectives

- Prepare a practice PC safely before opening the case
- Remove and refit RAM, storage cables or a GPU under supervision
- Use ESD controls and a check-before-power routine

Key Vocabulary

English	中文	Definition
ESD	静电释放	Static electricity damaging components
Anti-static wrist strap	防静电腕带	Wristband grounded to prevent ESD
Standoff	铜柱	Spacer holding motherboard in case
DIMM slot	内存插槽	RAM insertion slot
M.2 slot	M.2插槽	Slot for NVMe SSDs
PCIe retention latch	PCIe固定卡扣	Lock that holds GPU in place

STARTER

"ESD Myth Busters" — True or False? (1) You can build a PC on carpet if careful. (2) Touching case metal discharges static. (3) You need special gloves for RAM. (4) Installing GPU while PSU plugged in is safe.

"静电防护辟谣" — 判断对错：(1) 只要小心，可以在地毯上组装电脑。(2) 触摸机箱金属可以释放静电。(3) 拆装内存需要专用手套。(4) 电源插着时安装显卡是安全的。

Worksheet

1. T/F / 对/错：_____
2. T/F / 对/错：_____
3. T/F / 对/错：_____
4. T/F / 对/错：_____

MAIN 1

Instructor demonstration: safe case access — Watch, ask questions and complete the safety checklist before touching any component.

老师演示：安全打开机箱 — 先观察、提问并完成安全检查清单，再接触任何部件。

- ☐ Shut down through the operating system
- ☐ Switch off and unplug power
- ☐ Press the power button briefly to discharge
- ☐ Move to the ESD-safe work area
- ☐ Photograph cable positions
- ☐ Handle boards by the edges

Never open a power-supply unit. Do not remove a CPU or cooler in this student activity. / 绝不拆开电源内部。本次学生活动不拆卸CPU或散热器。

MAIN 2

Supervised upgrade stations — Rotate through the stations assigned by your instructor. One student handles the part, one reads the checklist and one records evidence. Swap roles.

监督下的升级实践站 — 按老师安排轮换实践站。一名学生操作部件，一名朗读检查清单，一名记录证据，然后交换角色。

Station	Safe action	Success check
RAM	Release latches; align notch; press evenly	Both latches closed
Storage	Identify SATA data/power or teacher-prepared M.2	No loose cable; screw secure
GPU	Release screw and PCIe latch; support the card	Card level; latch and screw secure

Teacher check before power / 通电前老师检查: ☐ no loose parts ☐ cables clear of fans ☐ side panel fitted ☐ tools removed

My evidence

Station completed / 完成的实践站: _____

One safety action I performed / 我执行的一项安全操作: _____

Teacher check initials / 老师确认: _____

REFLECT

"Build Confidence" — Rate your confidence 1–5 for: (a) Opening a PC, (b) Installing RAM, (c) Installing GPU.

"组装信心" — 为以下项目打分1–5: (a) 打开机箱, (b) 安装内存, (c) 安装显卡。

Reflection Space

Task	Confidence
Open case	_____
Install RAM	_____
Install GPU	_____

Scariest moment / 最紧张时刻: _____

At-home plan / 在家计划: _____

VISUAL SUMMARY

Safe component-handling loop

POWER DOWN

Shut down, switch off, unplug / 关机、断电、拔线

DISCHARGE

Press power button; use ESD control / 放电并做好静电防护

CHECK

Photo, label, release latch / 拍照、标记、松开卡扣

FIT + TEST

Never force; teacher checks before power / 不要硬插; 通电前老师检查

Never open a PSU / 绝不拆开电源内部

Lesson 6

AI Fundamentals & Model Types / 人工智能基础概念与类型

Suggested timing: 8 + 20 + 24 + 8 = 60 min

Learning Objectives

- Explain AI, machine learning, training and inference in simple terms
- Distinguish general, specialist, text and multimodal model uses
- Explain why model output must be checked

Key Vocabulary

English	中文	Definition
AI	人工智能	Computer systems that perform tasks associated with intelligent behaviour
Machine Learning	机器学习	AI that learns from data
Deep Learning	深度学习	ML using neural networks
LLM	大语言模型	AI trained on text to generate language
Parameters	参数	Internal variables learned during training
Training	训练	Teaching a model with data
Inference	推理	Using a trained model to predict
Token	词元	Unit of text (word or subword)
Hallucination	幻觉	Plausible-looking AI output that may be false or unsupported

STARTER

"AI or Not AI?" — Stand up if AI, sit if not: (1) Calculator solving 234×567 , (2) Spotify recommending songs, (3) Robot vacuum mapping, (4) Thermostat at 22°C, (5) ChatGPT writing an essay.

"是不是AI?" — 认为是AI的站起来，不是的坐着：(1) 计算器，(2) Spotify推荐，(3) 扫地机器人，(4) 恒温器，(5) ChatGPT。

Worksheet

Scenario	AI?	Why?
Calculator	_____	_____
Spotify	_____	_____
Robot vacuum	_____	_____
Thermostat	_____	_____
ChatGPT	_____	_____

MAIN 1

Model card comparison — Use the six current model cards supplied by your instructor. Do not guess unpublished parameter counts. Compare each model's inputs, outputs, local/cloud location, approximate hardware need and best-fit task.

模型卡比较 — 使用老师提供的六张当前模型卡。不要猜测未公开的参数量。比较每个模型的输入、输出、本地/云端位置、大致硬件需求和最适合的任务。

Model evidence table

Model	Text, image, audio?	Local or cloud	Best-fit task + evidence
Card A	_____	_____	_____
Card B	_____	_____	_____

Card C

Model names and services change quickly. Record the date printed on each card. / 模型名称和服务变化很快，请记录每张卡片上的日期。

MAIN 2

Tokenization Hands-On — Visit platform.openai.com/tokenizer (or offline demo). Type sentences and observe token splits.

分词实操 — 访问tokenizer（或使用老师提供的离线演示）。输入句子，观察词元拆分。

Try: "Hello AI Club", "Artificial Intelligence", "🚀💡💻", your full name.

尝试: "Hello AI Club"、"Artificial Intelligence"、表情符号、你的全名。

Token Count

Sentence	Tokens	Surprise?
Hello AI Club	_____	_____
Artificial Intelligence	_____	_____
🚀💡💻	_____	_____
My name	_____	_____

Discussion / 讨论: Why does "Artificial Intelligence" use more tokens than "AI"?

REFLECT

"AI Definition in My Own Words" — Write a one-paragraph explanation of AI for a 10-year-old. Include at least three vocabulary words.

"用我自己的话定义AI" — 用一段话向10岁小朋友解释AI。至少使用三个今天学到的词汇。

Reflection Space

VISUAL SUMMARY

From data to an answer

TRAINING DATA

Examples used during training / 训练阶段使用的示例

MODEL

Learns statistical patterns / 学习统计模式

PROMPT

Your current input / 你当前的输入

OUTPUT

A prediction—not guaranteed truth / 预测结果，并非保证正确

Verify important claims with reliable evidence / 重要信息要用可靠证据核实

Lesson 6 Notes

第6课笔记

Use this page for commands, test evidence, diagrams or questions to revisit. / 可记录命令、测试证据、图示或待解决问题。

Lesson 7

AI Ethics & School-Appropriate Use / 学校场景下的AI伦理与规范

Suggested timing: 8 + 20 + 24 + 8 = 60 min

Learning Objectives

- Identify ethical risks of AI in education
- Establish personal guidelines for AI-assisted learning
- Understand plagiarism, hallucination, and data privacy

Key Vocabulary

English	中文	Definition
Academic Integrity	学术诚信	Honest and responsible scholarship
Plagiarism	抄袭	Presenting others' work as your own
Hallucination	幻觉	Confident false output from AI
Bias	偏见	Systematic unfairness in AI output
Data Privacy	数据隐私	Protecting personal information
Prompt Injection	提示注入	Malicious input to manipulate AI
Critical Thinking	批判性思维	Evaluating information objectively

STARTER

“The essay dilemma” — Alex asks AI to write an entire essay and submits it with only small edits. In groups, discuss: What evidence of Alex's learning is now missing? Who owns the ideas and words? How could Alex use AI while still showing genuine learning?

“作文难题” — Alex让AI写完整篇作文，只做少量修改后提交。小组讨论：哪些学习证据消失了？观点和语言属于谁？Alex怎样使用AI才能仍然展示真实学习？

A better workflow

Plan myself / 自己规划 → _____ → Verify / 核实 → Rewrite / 自己重写 → Disclose / 说明使用情况

MAIN 1

AI Policy Drafting Workshop — Your school has no AI policy. Draft a 5-rule "Student AI Use Code" with "Why" explanations.

AI政策起草工作坊 — 你的学校目前没有AI政策。分组起草5条规则的“学生AI使用守则”，每条必须有“为什么”的解释。

STUDENT AI USE CODE

Class: _____ Date / 日期: _____

Rule 1 / 规则1: _____

Why / 为什么: _____

Rule 2 / 规则2: _____

Why / 为什么: _____

Rule 3 / 规则3: _____

Why / 为什么: _____

Rule 4 / 规则4: _____

Why / 为什么: _____

Rule 5 / 规则5: _____

Why / 为什么: _____

MAIN 2

Spot the unsupported claim — Your instructor provides three short AI answers and a small source pack. One answer contains a confident claim that the sources do not support. Highlight the claim, cite the evidence and rewrite it accurately.

找出缺乏证据的说法 — 老师提供三段AI回答和一组简短资料。其中一段包含资料不支持但语气肯定的说法。标出该说法，引用证据并准确改写。

Claim check

Claim / 说法: _____

Evidence source + line / 证据来源及行号: _____

Accurate rewrite / 准确改写: _____

REFLECT

"My AI Promise" — Write a personal pledge for how YOU will use AI this school year. Sign it.

"我的AI承诺" — 写下你这学年使用AI的个人承诺。签名。

Reflection Space

I, _____ (name / 姓名), pledge to:

1. _____

2. _____

3. _____

Signed / 签名: _____ Date / 日期: _____

VISUAL SUMMARY

Responsible AI learning loop

ASK

Use a clear learning purpose / 明确学习目的

CHECK

Verify facts, bias and missing context / 核查事实、偏见与缺失语境

CREATE

Think, revise and add your own judgement / 独立思考、修改并加入判断

DISCLOSE

Explain how AI helped / 说明AI如何提供帮助

No personal data, passwords, private work or undisclosed copying / 不输入个人数据、密码、私密内容，不隐瞒照抄

Lesson 8

OpenClaw Agent Installation & Setup / OpenClaw智能体安装与初始化

Suggested timing: 7 + 28 + 18 + 7 = 60 min

Learning Objectives

- Understand what OpenClaw is and its ecosystem
- Install the OpenClaw CLI agent on Ubuntu
- Run initial setup and verify installation

Key Vocabulary

English	中文	Student-friendly meaning
Agent	智能体	Software that can use a model and approved tools to complete tasks
CLI	命令行界面	A text interface for running commands
Gateway	网关	The OpenClaw service that connects sessions, models, channels and nodes
Control UI	控制界面	The browser dashboard used to inspect and operate OpenClaw
Channel	通道	A messaging connection such as Telegram
Pairing	配对	An explicit approval step for access
Scope	权限范围	A limit on what an approved account or device may do

STARTER

"What is OpenClaw?" — OpenClaw is an AI orchestration agent. Draw a diagram: Your Laptop → OpenClaw → [What can it connect to?]. Brainstorm in pairs.

什么是OpenClaw? — OpenClaw是一个AI编排智能体。画一个图：你的笔记本 → OpenClaw → [它能连接到什么?]。两人一组头脑风暴。

Worksheet

[My Laptop / 我的笔记本] → [OpenClaw] → [_____]
↓
[_____]

MAIN 1

Install and onboard OpenClaw — The instructor checks the official documentation and prepares the network before class. Use only the assigned ECA machine. OpenClaw currently requires a supported Node.js runtime; the official installer checks prerequisites and starts onboarding.

安装并初始化OpenClaw — 老师在课前核对官方文档并准备网络。仅使用指定的ECA电脑。OpenClaw目前需要受支持的Node.js运行环境；官方安装程序会检查前提条件并启动初始化。

Teacher verification / 老师确认： official page is <https://docs.openclaw.ai/start/getting-started>; no student enters a personal provider key.

```
node --version

# Run only after the instructor confirms the official URL:
curl -fsSL https://openclaw.ai/install.sh | bash

openclaw --version
# The installer normally starts onboarding. To return later:
openclaw setup
```

Installation record

Node.js version / Node.js版本: _____

OpenClaw version / OpenClaw版本: _____

Onboarding route used / 初始化方式: ☐ local model / 本地模型 ☐ teacher-managed provider / 老师管理的服务

Do not record API keys or tokens. / 不要记录API密钥或令牌。

MAIN 2

Gateway health and Control UI — Use read-only checks first. Open the dashboard and identify the gateway, model and session status. Run a repair command only after the instructor has reviewed the diagnosis.

网关健康与控制界面 — 先使用只读检查。打开控制面板，识别网关、模型和会话状态。只有老师查看诊断结果后才能运行修复命令。

```
openclaw gateway status
openclaw dashboard
openclaw status --deep
openclaw doctor
```

```
# Teacher approval required / 需要老师批准:
openclaw doctor --fix
```

Health check

Gateway status / 网关状态: _____

Control UI opened? / 控制界面已打开? ☐ Yes / 是

One doctor finding / 一项诊断结果: _____

Action taken / 采取的行动: ☐ none / 无 ☐ teacher-approved repair / 老师批准的修复

REFLECT

"OpenClaw First Impressions" — What does OpenClaw do that a web browser cannot? What confused you? Write one question for next lesson.

"OpenClaw初印象" — OpenClaw能做什么，是浏览器做不到的？安装过程中哪里困惑？为下节课写一个问题。

Reflection Space

Browser vs OpenClaw / 浏览器 vs OpenClaw:

Confusing part / 困惑之处:

Question for next lesson / 下节课的问题:

VISUAL SUMMARY

OpenClaw in one picture / OpenClaw系统概览

CONTROL UI / CLI

Human gives instructions / 人类发出指令

GATEWAY

Routes sessions, tools and channels / 调度会话、工具和通道

MODEL

Cloud or local inference / 云端或本地模型

CHANNELS / NODES

Only approved connections / 仅允许已批准的连接

Permissions limit what an agent may do / 权限决定智能体可以做什么

第8课笔记

Lesson 9

OpenClaw Channels, Pairing & Permissions / OpenClaw通道、配对与权限

Suggested timing: 8 + 22 + 23 + 7 = 60 min

Learning Objectives

- Explain the difference between a channel, device, node, role and scope
- Observe a teacher-controlled Telegram channel setup
- Review pairing requests without exposing credentials

Key Vocabulary

English	中文	Definition
Bot Token	机器人令牌	Secret key for Telegram bot
Polling	轮询	Checking for new messages
Scope	权限范围	Level of access
QR Pairing	二维码配对	Scanning QR to link devices
Role	角色	Function of device in network

STARTER

"Channel Brainstorm" — List all ways you communicate digitally. For each, discuss: Could an AI agent read/respond through this channel? Privacy implications?

"通道头脑风暴" — 列出所有数字通信方式。对每个讨论：AI智能体能通过这个渠道阅读和回复吗？隐私影响是什么？

Worksheet

Channel	AI Possible? / AI可行?	Privacy Risk
WeChat	_____	_____
Email	_____	_____
Telegram	_____	_____
School LMS	_____	_____

MAIN 1

Teacher-controlled Telegram channel — Students do not create bots or receive the bot token. The instructor adds the ECA demonstration bot, then the class inspects status and completes one DM pairing example.

老师控制的Telegram通道 — 学生不创建机器人，也不会获得机器人令牌。老师添加ECA演示机器人，全班查看状态并完成一次私聊配对示例。

```
# Instructor enters the token privately / 老师私下输入令牌:
openclaw channels add --channel telegram

# Safe class checks / 课堂安全检查:
openclaw channels list
openclaw channels status --channel telegram --probe
openclaw pairing list telegram

# Approve only the exact displayed code after identity check:
openclaw pairing approve telegram <CODE>
```

Never write or screenshot a bot token. Pairing a sender allows access; the instructor must confirm the requester. / 不要写下或截图机器人令牌。配对会授予访问权限；必须由老师确认请求者。

Pairing decision

Requested channel / 请求通道: _____
 Identity checked how? / 如何核对身份? _____
 Decision / 决定: ☐ approve / 批准 ☐ reject / 拒绝 Reason / 理由: _____

MAIN 2

Device and permission review — Use printed sample requests or a teacher demonstration. Compare requested role and scopes before making a decision. If the instructor demonstrates mobile pairing, use a limited setup code.

设备与权限审核 — 使用打印的示例请求或老师演示。作出决定前比较所请求的角色和权限范围。若老师演示移动配对，应使用受限配对码。

```
openclaw devices list
openclaw qr --limited          # teacher demonstration / 老师演示
openclaw devices list        # obtain the exact requestId
openclaw devices approve <requestId>
openclaw devices remove <deviceId>
```

Least-privilege check

Request	Needed for task?	Approve, reject or ask?
Read status	_____	_____
Administrator scope	_____	_____
Unknown device	_____	_____

REFLECT

"My OpenClaw Network" — Draw your current OpenClaw setup. What channels and devices do you have? What would you add at home?

"我的OpenClaw网络" — 画出你当前的OpenClaw设置。你有哪些通道和设备？如果这是家庭服务器，你会添加什么？

Reflection Space

VISUAL SUMMARY

Pairing is an access decision

REQUEST

A person/device asks to connect / 人或设备请求连接

REVIEW

Check identity, role and scopes / 核对身份、角色与权限

APPROVE or REJECT

Teacher/operator decides / 老师或管理员决定

REVIEW LATER

List, rename, rotate or remove / 之后可查看、重命名、轮换或移除

A bot token or setup code is a secret while valid / 机器人令牌或有效配对码属于秘密

Lesson 10

OpenClaw Maintenance & Troubleshooting / OpenClaw日常维护与故障排查

Suggested timing: 7 + 23 + 23 + 7 = 60 min

Learning Objectives

- Read and interpret OpenClaw logs
- Apply updates safely
- Diagnose and fix common issues using doctor --fix

Key Vocabulary

English	中文	Student-friendly meaning
Log	日志	A time-ordered record of system events
Status	状态	A summary of whether a component is available and healthy
Probe	探测	A live check that tests whether a service responds
Diagnosis	诊断	Evidence-based identification of a problem
Repair	修复	A change intended to correct a diagnosed problem
Dry run	模拟运行	A preview that does not apply the planned change
Rollback	回滚	Returning to a known working version or configuration

STARTER

"Bug Hunt Warm-Up" — Your teacher will display three error messages (one fake, two real). Vote on which is the real OpenClaw error. Discuss: What clues help you identify real vs fake errors?

"找虫热身" — 老师将展示三条错误信息（一条假的，两条真的）。投票哪条是真的OpenClaw错误。讨论：什么线索帮助你区分真假错误？

Worksheet

Error A / 错误A: Real / Fake / 真 / 假 | Why? / 为什么? _____

Error B / 错误B: Real / Fake / 真 / 假 | Why? / 为什么? _____

Error C / 错误C: Real / Fake / 真 / 假 | Why? / 为什么? _____

MAIN 1

Log analysis lab — Use OpenClaw's supported log command rather than guessing a file path. Search the displayed lines for a timestamp, level, component and message. Redact secrets before copying any example.

日志分析实验 — 使用OpenClaw支持的日志命令，不要猜测文件路径。在显示内容中寻找时间戳、级别、组件和消息。复制示例前先遮盖秘密信息。

```
openclaw logs --limit 50 --plain
openclaw logs --follow          # Ctrl+C to stop / 按Ctrl+C停止
openclaw channels logs --channel all
openclaw channels status --channel telegram --probe
```

Log detective

Timestamp / 时间戳: _____ Level / 级别: _____

Component / 组件: _____ Message in my own words / 用自己的话解释: _____

Secret or personal data removed? / 已移除秘密或个人数据? ☐ Yes / 是

MAIN 2

Troubleshooting scenarios — Work from evidence. Change only one thing at a time and record the result. The instructor decides whether any repair or update is applied.

故障排查场景 — 依据证据开展工作。一次只更改一项并记录结果。是否执行修复或更新由老师决定。

```
# 1. Broad health / 整体健康
openclaw gateway status
openclaw status --deep

# 2. Read-only diagnosis / 只读诊断
openclaw doctor

# 3. Preview updates / 预览更新
openclaw update status
openclaw update --dry-run

# Teacher approval required / 需要老师批准
openclaw doctor --fix
```

Troubleshooting record

Symptom	Evidence	One action	Result
_____	_____	_____	_____
_____	_____	_____	_____

REFLECT

"Maintenance Mindset" — Why is regular maintenance better than fixing things only when they break? How often should you check logs? Write a weekly maintenance checklist.

"维护心态" — 为什么定期维护比坏了再修更好？你应该多久检查一次日志？写一份每周维护检查清单。

My Weekly Checklist

- ☐ _____
- ☐ _____
- ☐ _____
- ☐ _____

VISUAL SUMMARY

Troubleshooting ladder

1 OBSERVE

Write the exact symptom / 记录准确症状

2 STATUS

gateway status • status --deep / 检查状态

3 LOGS

logs --limit 50 --plain / 查看日志

4 DIAGNOSE

doctor first; fix only with approval / 先诊断，经批准再修复

Change one thing, test, and record the result / 一次只改一项，测试并记录结果

Lesson 11

Local AI Environment Setup / 本地AI运行环境搭建

Suggested timing: 7 + 23 + 23 + 7 = 60 min

Learning Objectives

- Install Ollama on Ubuntu
- Pull and run a lightweight model
- Understand CPU vs GPU inference

Key Vocabulary

English	中文	Definition
Ollama	Ollama	Tool to run LLMs locally
Model weights	模型权重	Trained parameters file
Quantization	量化	Reducing precision for speed/size
Inference	推理	Running model to get output
CPU inference	CPU推理	Running AI on processor
GPU inference	GPU推理	Running AI on graphics card
Context window	上下文窗口	How much text model remembers

STARTER

"Why Local AI?" — Discuss in groups: (1) What if you have no internet? (2) What if cloud AI is down? (3) What if you don't want to send private data online? List three situations where local AI is better.

"为什么本地AI?" — 分组讨论: (1) 如果没网怎么办? (2) 如果云端AI宕机怎么办? (3) 如果你不想把私人数据发到网上怎么办? 列出三种本地AI优于云端AI的场景。

Worksheet

Situation 1 / 场景1: _____

Situation 2 / 场景2: _____

Situation 3 / 场景3: _____

MAIN 1

Verify the instructor-prepared Ollama environment — The instructor may install Ollama and pre-download models before class. If installation is part of your session, verify the official URL first.

检查老师准备的Ollama环境 — 老师可能在课前安装Ollama并预下载模型。若本节课包含安装, 请先核对官方网址。

```
# Official installer; teacher approval required / 官方安装程序, 需老师批准
curl -fsSL https://ollama.com/install.sh | sh
```

```
ollama --version
systemctl is-active ollama
ollama list
```

Environment check

Ollama version / 版本: _____ Service active? / 服务已启动? ☐ Yes / 是

Preloaded model(s) / 已预载模型: _____

Do not download several large models during the lesson unless instructed. / 除非老师要求, 不要在课堂上下载多个大型模型。

MAIN 2

Your first local comparison — Run one preloaded small multilingual model. Use the same three prompts as your partner, record speed and check one factual answer.

首次本地模型比较 — 运行一个预载的小型多语言模型。与同伴使用相同的三个提示，记录速度并核实一个事实回答。

```
ollama list
ollama run qwen3:0.6b

# Try in the chat / 在对话中尝试:
# 1. Explain one school topic in simple English.
# 2. Explain the same topic in 简体中文.
# 3. Ask one question whose answer you can verify.
# Exit / 退出: /bye
```

Local model evidence

Model / 模型: _____ First response time / 首次响应时间: _____

Useful strength / 有用优点: _____

Limitation or error / 局限或错误: _____

How I verified one claim / 我如何核实一项说法: _____

REFLECT

"Local vs Cloud" — Compare your experience with local Ollama vs using ChatGPT online. What are three pros and three cons of local AI?

"本地 vs 云端" — 对比本地Ollama和在线使用ChatGPT的体验。本地AI的三个优点和三个缺点是什么？

Reflection Space

Local AI Pros	Local AI Cons
1. _____	1. _____
2. _____	2. _____
3. _____	3. _____

VISUAL SUMMARY

Local AI data path

MODEL DOWNLOAD

Internet needed once / 首次下载需要网络

OLLAMA

Runs model on this computer / 在本机运行模型

PROMPT

Stays local when no external tools are used / 不调用外部工具时留在本机

OUTPUT

Speed depends on model + hardware / 速度取决于模型与硬件

Local does not automatically mean accurate or safe / 本地运行并不等于自动准确或安全

Lesson 12

Lightweight Model Selection & Optimization / 轻量化模型选择与优化部署

Suggested timing: 7 + 23 + 23 + 7 = 60 min

Learning Objectives

- Compare model sizes and quantization levels
- Select appropriate models for specific hardware
- Optimize performance with context window tuning

Key Vocabulary

English	中文	Definition
Quantization (Q4, Q8)	量化 (Q4、Q8)	Bit precision reduction
Parameter count	参数量	Number of weights in model
VRAM	显存	Video RAM for GPU inference
System RAM	系统内存	Regular RAM for CPU inference
Temperature	温度	Creativity randomness
Top-p	核采样	Nucleus sampling threshold

STARTER

Model speed dating — Match instructor-supplied model cards to devices and tasks. Treat RAM estimates as approximate: the exact requirement depends on quantisation, context length, runtime and other processes.

模型快速配对 — 将老师提供的模型卡与设备和任务匹配。内存估算只是近似值：实际需求取决于量化、上下文长度、运行环境和其他进程。

One justified match

Device / 设备: _____ Model / 模型: _____

Evidence / 证据 (size, RAM, task, language): _____

MAIN 1

Hardware-model comparison — Use two teacher-preloaded models. Keep the prompt, machine and timing method the same. Quality is judged with a simple checklist, not speed alone.

硬件与模型比较 — 使用两个由老师预载的模型。保持提示、电脑和计时方法一致。质量要用简明检查表判断，不能只看速度。

```
free -h
ollama list
ollama show qwen3:0.6b
ollama show llama3.2:1b

time ollama run qwen3:0.6b "Explain photosynthesis in 60 words."
time ollama run llama3.2:1b "Explain photosynthesis in 60 words."
```

Fair comparison

Model	Time	Accurate?	Clear?	Followed 60 words?
qwen3:0.6b	_____	_____	_____	_____
llama3.2:1b	_____	_____	_____	_____

MAIN 2

Modelfile customisation — Build a small tutor variant. Change one parameter only, test the same prompt before and after, and describe the difference without claiming that one setting is always best.

自定义Modelfile — 创建一个小型导师模型。一次只改变一个参数，在更改前后使用相同提示，并描述差异，不要声称某个设置永远最好。

```
nano ~/AI_Club/Modelfile

FROM qwen3:0.6b
PARAMETER temperature 0.4
PARAMETER num_ctx 2048
SYSTEM "You are a careful bilingual study coach. Explain simply, ask one checking question, and say when you are uncertain."

cd ~/AI_Club
ollama create my-study-coach -f Modelfile
ollama run my-study-coach
```

One-change test

Parameter changed / 更改的参数: _____ From / 从: _____ To / 到: _____
Observed difference / 观察到的差异: _____

REFLECT

"My Model Recommendation" — Write a recommendation for a student with a 10-year-old laptop (4GB RAM, no GPU). Which model should they use? What settings?

"我的模型推荐" — 为一位使用10年前旧笔记本（4GB内存，无独显）的学生写推荐。该用哪个模型？什么设置？

Reflection Space

Recommended model / 推荐模型: _____
Why / 为什么: _____
Settings to change / 需要调整的设置: _____

VISUAL SUMMARY

Model choice is a trade-off

SMALLER

Faster • less RAM • lower capability / 更快 • 更省内存 • 能力较低

LARGER

Often stronger • slower • more RAM / 通常更强 • 更慢 • 更占内存

SHORT CONTEXT

Lower memory use / 更省内存

LONG CONTEXT

More text • more memory and time / 更多文本 • 更多内存与时间

Choose the smallest model that meets the task / 选择能完成任务的最小模型

Lesson 13

Hands-On Local AI Projects / 本地AI实际应用与项目实践

Suggested timing: 6 + 25 + 22 + 7 = 60 min

Learning Objectives

- Build a practical text generation pipeline
- Create a simple translation assistant
- Export and save conversation logs

Key Vocabulary

English	中文	Definition
Pipeline	流水线	Sequence of processing steps
Prompt engineering	提示工程	Crafting effective inputs for AI
Few-shot prompting	少样本提示	Giving examples in the prompt
System prompt	系统提示词	Background instructions for AI
Conversation history	对话历史	Record of past exchanges

STARTER

"Prompt Magic" — Everyone types the same basic prompt: "Write about dogs." Then try: "Write a persuasive paragraph about why dogs make better pets than cats, aimed at 12-year-olds, using at least one statistic." Compare outputs. What changed?

"提示词魔法" — 所有人都输入相同的基础提示："写关于狗的内容。"然后尝试："写一段有说服力的文字，说明为什么狗比猫更适合做宠物，面向12岁读者，至少使用一个统计数据。"对比输出。什么变了？

Worksheet

Basic output quality (1-5) / 基础输出质量：_____

Detailed output quality (1-5) / 详细输出质量：_____

One word that made the biggest difference / 影响最大的一个词：_____

MAIN 1

Build a Study Buddy — Create a custom model that helps you study any subject. Test it with three questions.

构建学习伙伴 — 创建一个帮助你学习任意科目的自定义模型。用三个问题测试它。

```
nano ~/AI_Club/study-buddy.modelfile

# Example content:
FROM llama3.2
PARAMETER temperature 0.3
PARAMETER num_ctx 4096
SYSTEM ""You are a patient study tutor. When asked a question:
1. Explain the concept simply
2. Give a real-world analogy
3. Ask the student a follow-up question
4. Never give the full answer immediately.""

ollama create study-buddy -f ~/AI_Club/study-buddy.modelfile
ollama run study-buddy
```

Study Buddy Test

Question	Analogy?	Follow-up?	Helpful?
1. Why does ice float?	_____	_____	_____
2. Pythagorean theorem?	_____	_____	_____

MAIN 2

Translation and summarisation pipeline — Use only the instructor's non-personal practice article. Save the prompt, raw model output and your checked revision as three separate files.

翻译与摘要流程 — 仅使用老师提供的不含个人信息的练习文章。将提示、模型原始输出和自己核查修改后的版本分别保存为三个文件。

```
nano ~/AI_Club/source_article.txt
nano ~/AI_Club/summary_prompt.txt

# Put the task instructions first, then paste the practice article.
cat ~/AI_Club/summary_prompt.txt ~/AI_Club/source_article.txt \
| ollama run qwen3:0.6b \
> ~/AI_Club/model_draft.txt

cp ~/AI_Club/model_draft.txt ~/AI_Club/checked_final.txt
nano ~/AI_Club/checked_final.txt
```

Human review log

One fact checked / 核实的一项事实: _____

One language improvement / 一项语言改进: _____

One model limitation / 一项模型局限: _____

REFLECT

"Project Showcase Prep" — You will demo one of today's projects in Lesson 15. Which project are you most proud of? What would you improve with more time?

"项目展示准备" — 你将在第15课展示今天的某个项目。你最自豪的是哪个？如果有更多时间，你会怎么改进？

Reflection Space

Project to showcase / 要展示的项目: _____

Why proud / 为什么自豪: _____

Improvement idea / 改进想法: _____

VISUAL SUMMARY

A reliable AI project pipeline**INPUT**

Purpose + safe source text / 目标+安全的原文

INSTRUCTIONS

Role + task + limits + format / 角色+任务+限制+格式

MODEL OUTPUT

Draft, not final truth / 草稿，不是最终事实

HUMAN REVIEW

Verify • edit • save evidence / 核实 • 修改 • 保存证据

Keep the original text and your revised final version / 保留原文和自己修改后的最终版本

Lesson 14

Responsible Use of School-Approved Cloud AI / 负责任地使用学校批准的云端AI

Suggested timing: 8 + 22 + 23 + 7 = 60 min

Learning Objectives

- Decide what information must not be sent to a cloud AI service
- Use teacher-provided access and compare outputs responsibly
- Apply data minimisation, verification and disclosure

Key Vocabulary

English	中文	Student-friendly meaning
Cloud AI	云端AI	An AI service that processes requests on remote servers
PII	个人身份信息	Information that can identify a person
Data minimisation	数据最小化	Sharing only the information required for the task
Data retention	数据保留	How long a service may keep submitted information
Age requirement	年龄要求	A service rule limiting who may create or use an account
Verification	核实	Checking an output against trustworthy evidence
Disclosure	使用说明	Explaining how AI contributed to the work

STARTER

"Terms of Service Speed Read" — Your teacher will display three short excerpts from AI service Terms of Service. Highlight in red anything that worries you about privacy. Highlight in green anything that protects you.

"服务条款速读" — 老师将展示三条AI服务条款的摘录。用红色标出让你担忧隐私的内容。用绿色标出保护你的内容。

Worksheet

Excerpt A / 摘录A: Red flag / 红旗: _____ Green flag / 绿旗: _____
Excerpt B / 摘录B: Red flag / 红旗: _____ Green flag / 绿旗: _____
Excerpt C / 摘录C: Red flag / 红旗: _____ Green flag / 绿旗: _____

MAIN 1

Teacher-managed comparison — Do not create a personal account in this lesson. Use the approved classroom access or watch the instructor demonstration. Send the same non-personal prompt to two available services and compare usefulness, evidence and uncertainty.

老师管理的模型比较 — 本节课不要创建个人账户。使用经批准的课堂访问方式或观看老师演示。向两个可用服务发送相同且不含个人信息的提示，并比较实用性、证据和不确定性。

Services, availability, age rules and terms can change. The instructor confirms approved access on the teaching date. / 服务、可用性、年龄规则和条款可能变化，老师须在授课当天确认批准的访问方式。

Same-prompt comparison

Service	Useful detail	Evidence or citation	Uncertainty stated?
A	_____	_____	_____
B	_____	_____	_____

MAIN 2

Privacy filter challenge — Rewrite each risky prompt so it contains only the minimum information needed. Never submit the original risky prompt.

隐私过滤挑战 — 改写每个有风险的提示，使其只包含完成任务所需的最少信息。不要提交原始的高风险提示。

Remove → Generalise → Ask

Risky detail	Safer replacement
Student's full name and report	Anonymous sample paragraph
Photo showing faces and ID cards	Describe the non-identifying feature
Password, API key or bot token	Use the placeholder <SECRET>; never send the real value

My safer prompt / 我的安全提示: _____

REFLECT

"My Cloud AI Safety Pledge" — Write three personal rules for using cloud AI that you will follow this school year.

"我的云端AI安全承诺" — 写下你这学年使用云端AI的三条个人规则。

Reflection Space

Rule 1 / 规则1: _____

Rule 2 / 规则2: _____

Rule 3 / 规则3: _____

VISUAL SUMMARY

Before sending data to cloud AI

PAUSE

Do I have permission? / 我有权限吗?

REMOVE

Names, IDs, faces, passwords, private files / 姓名、证件、面部、密码、私密文件

MINIMISE

Share only what the task needs / 仅提供任务所需信息

VERIFY

Check output and cite real sources / 核查输出并引用真实来源

Use only school-approved services and teacher-provided access / 仅使用学校批准的服务和老师提供的访问方式

Lesson 15

Capstone Project & Showcase / AI工具综合项目与成果展示

Suggested timing: 5 + 25 + 23 + 7 = 60 min

Learning Objectives

- Combine Ubuntu, OpenClaw, local AI, and cloud AI skills into one project
- Present technical work clearly to peers
- Provide and receive constructive feedback

Key Vocabulary

English	中文	Definition
Capstone	毕业设计/顶点项目	Culminating project demonstrating skills
Demo	演示	Live demonstration of working project
Q&A	问答	Question and answer session
Feedback	反馈	Constructive criticism
Portfolio	作品集	Collection of your work

STARTER

"Project Pitch Roulette" — Each student has 60 seconds to pitch their project idea to a partner. The partner asks one question. Switch roles. Refine your pitch.

"项目提案轮盘" — 每位学生有60秒向搭档陈述项目想法。搭档提一个问题。交换角色。完善你的陈述。

Pitch Draft

My project name / 项目名称: _____

In one sentence / 一句话描述: _____

One question I was asked / 被问到的一个问题: _____

My improved pitch / 改进后的陈述: _____

MAIN 1

Final build and test — Improve one project started earlier. Aim for a simple, functioning and well-explained result. Teacher approval is required for any external channel, account or hardware change.

最终制作与测试 — 改进之前开始的一个项目。目标是简单、可运行并解释清楚。任何外部通道、账户或硬件更改都需要老师批准。

Project option	Minimum success
Local Study Coach	Custom Modelfile + three tested prompts + one corrected error
OpenClaw Health Reporter / OpenClaw健康报告器	Safe script using status, gateway status and doctor; no automatic repair
Responsible AI Guide	One-page bilingual guide with privacy, verification and disclosure examples
Model Comparison	Fair same-prompt test with evidence and a justified recommendation
Hardware Upgrade Plan	Current-spec proposal with compatibility, safety and budget reasoning

25-minute build plan / 25分钟制作计划

My project / 我的项目: _____

Must finish / 必须完成: _____

Test evidence / 测试证据: _____

Safety/privacy check / 安全与隐私检查: ☐ complete / 已完成

MAIN 2

Gallery walk and peer review — Half the class presents while half visits, then swap. Each presenter gives a 90-second explanation and answers one question. Visitors complete two focused review cards.

作品巡展与同伴评价 — 一半学生展示，另一半参观，然后交换。每位展示者进行90秒说明并回答一个问题。参观者完成两张重点评价卡。

Peer review card

Project / 项目: _____ Reviewer / 评价者: _____

One thing that worked well / 一个做得好的地方: _____

One piece of evidence I saw / 我看到的一项证据: _____

One useful next step / 一个有用的下一步: _____

REFLECT

"Course Reflection & Next Steps" — Complete the final reflection. This will be kept in your portfolio.

"课程反思与下一步" — 完成最终反思。这将保存在你的作品集中。

Final Reflection

1. Proudest moment / 最自豪的时刻: _____

2. Hardest challenge / 最大的挑战: _____

3. Skill I will use again / 我会再次使用的技能: _____

4. What I want to learn next / 接下来想学什么: _____

5. One word for this course / 形容本课程的一个词: _____

Certificate of Completion / 结业证书

This certifies that _____ (Student Name / 学生姓名)

has completed the AI Club 15-lesson course. / 已完成AI俱乐部15节课程。

Date / 日期: _____ Instructor / 指导老师: _____

VISUAL SUMMARY

Capstone success formula

WORKS

A functioning demo or clear prototype / 可运行演示或清晰原型

EXPLAINS

Purpose, method and one challenge / 目标、方法和一个挑战

SAFE

No secrets or personal data / 无秘密或个人数据

REFLECTS

Evidence, feedback and next step / 证据、反馈和下一步

Simple + tested + well explained beats complicated + unfinished / 简单、经过测试、解释清楚，胜过复杂但未完成

Appendix A — Linux Command Cheat Sheet / Linux命令速查表

```
# NAVIGATION / 导航
pwd                # Current directory / 当前目录
cd ~               # Home / 主目录
cd ..              # Up one level / 返回上一级
ls -la             # Include hidden entries / 包含隐藏项目
tree               # Show a directory tree / 显示目录树

# FILES / 文件 — practise inside ~/AI_Club/sandbox
touch file.txt     # Create an empty file / 创建空文件
mkdir -p a/b/c     # Create nested directories / 创建嵌套目录
cp -i a b          # Copy with overwrite confirmation / 确认后复制
mv -i a b          # Move/rename with confirmation / 确认后移动或重命名
rm -i file.txt     # Delete one practice file with confirmation / 确认后删除练习文件
cat file.txt       # View text / 查看文本
nano file.txt      # Edit text / 编辑文本
head -n 5 file.txt # First five lines / 前五行
tail -n 5 file.txt # Last five lines / 后五行

# SYSTEM / 系统
sudo command       # Temporary administrator privilege / 临时管理员权限
sudo apt update    # Refresh package lists / 更新软件包列表
sudo apt install pkg # Install an approved package / 安装批准的软件包
htop               # Process monitor / 进程监控
df -h              # Disk space / 磁盘空间
free -h            # Memory use / 内存使用

# SEARCH + PERMISSIONS / 搜索与权限
grep "text" file.txt # Find text / 查找文本
find . -name "*.txt" # Find files below current directory / 在当前目录下查找文件
chmod +x script.sh   # Make a trusted script executable / 使可信脚本可执行
```

Appendix B — AI Vocabulary Glossary / AI词汇表

English	中文	Student-friendly meaning
Algorithm	算法	A clear set of steps for solving a problem
Neural Network	神经网络	A model of connected mathematical units that learns patterns from data
Training Data	训练数据	Examples used to help a model learn patterns
Fine-tuning	微调	Further training a model for a narrower task
Prompt	提示词	The instruction, question or context given to a model
Embedding	嵌入	A numerical representation that helps software compare meaning
Vector Database	向量数据库	A database designed to store and search numerical representations
RAG	检索增强生成	A method that retrieves sources before a model drafts an answer
Agent	智能体	Software that can use a model and approved tools to complete tasks
Multimodal	多模态	A system that can work with more than one data type, such as text and images

Appendix C — Hardware Component Guide

CPU / 中央处理器

Runs instructions. Check socket, cores, threads, generation and cooling support. / 执行指令；检查插槽、核心、线程、代数和散热支持。

RAM / 内存

Holds active data. Check DDR generation, capacity, speed and supported configuration. / 保存活动数据；检查DDR代数、容量、速度和受支持配置。

Storage / 存储

HDD, SATA SSD and NVMe SSD differ in speed, connector and price. Back up before replacement. / HDD、SATA SSD和NVMe SSD在速度、接口和价格上不同；更换前先备份。

GPU / 显卡

Handles graphics and parallel computation. Check slot, size, power connectors, PSU capacity and software support. / 处理图形和并行计算；检查插槽、尺寸、供电接口、电源容量和软件支持。

Motherboard / 主板

Connects components. Check CPU socket, chipset, RAM type, form factor and expansion slots. / 连接各部件；检查CPU插槽、芯片组、内存类型、板型和扩展槽。

PSU / 电源

Supplies power. Match total load, connectors, quality and efficiency. Never open the PSU enclosure. / 提供电力；匹配总功耗、接口、质量和效率；绝不拆开电源外壳。

Appendix D — Useful Links & Resources

Resource	Official URL	Use
Ubuntu Desktop	ubuntu.com/download/desktop	Current desktop LTS and installation guidance
Ubuntu 24.04 LTS images	releases.ubuntu.com/noble	School-approved 24.04 LTS image source
OpenClaw Getting Started	docs.openclaw.ai/start/getting-started	Current prerequisites, installer and onboarding
OpenClaw CLI	docs.openclaw.ai/cli	Current command reference
Ollama	ollama.com/download	Local model runtime
Ollama Library	ollama.com/library	Model cards, sizes and licences
VS Code	code.visualstudio.com	Code editor
LibreOffice	libreoffice.org	Office suite
Partnership on AI	partnershiponai.org	Responsible-AI resources

Technical verification date / 技术核对日期：2026-08-27. Software commands, model tags, service availability, age requirements and terms can change. The instructor must recheck official sources before each course run. / 软件命令、模型标签、服务可用性、年龄要求 and 条款可能变化；每次开课前老师都必须重新核对官方来源。

Appendix E — Final Skills Passport & Next Steps / 附录E：最终技能护照与下一步

Rate your confidence now, then support one rating with evidence from your work. / 评估你现在的信心，并用作品证据说明其中一项评分。

I can... / 我能……	Now (1–4)
work safely with a computer and identify key components / 安全使用电脑并识别主要部件	1 2 3 4
navigate Linux and manage practice files in the terminal / 使用Linux并在终端中管理练习文件	1 2 3 4
explain how an AI model learns patterns and why it can make mistakes / 解释AI模型如何学习模式及为何会出错	1 2 3 4
test an AI answer using evidence instead of trusting it automatically / 用证据检验AI回答，而不是自动相信	1 2 3 4
protect personal data, passwords, tokens and school information / 保护个人数据、密码、令牌和学校信息	1 2 3 4
run and compare an approved local AI model / 运行并比较经批准的本地AI模型	1 2 3 4
use OpenClaw status and diagnostic commands safely / 安全使用OpenClaw状态和诊断命令	1 2 3 4
plan, test and explain a small responsible-AI project / 规划、测试并解释一个负责任AI小项目	1 2 3 4

Evidence from my work / 我的作品证据

Skill / 技能： _____
Project, test or page that shows it / 证明它的项目、测试或页码： _____
What I can do now / 我现在能做到： _____
One responsible next step / 一个负责任的下一步： _____

Club reflection / 课程反思

The most useful habit I developed was... / 我养成的最有用习惯是……

Notes

General notes, questions and reminders / 通用笔记、问题与提醒

Project Notes & Ideas

Sketch ideas, workflows and improvements / 记录项目想法、流程与改进

Command Notes

Useful commands, flags and safe troubleshooting steps / 实用命令、参数与安全排错步骤

Portfolio Evidence & Next Steps

Record evidence you may want to show later / 记录以后可能展示的作品证据与下一步

NAS JIAXING

AI CLUB

Build • Understand • Create • Verify

After-School ECA • Student Coursebook v1.2